

UVINDU HEWAGE

Associate Software Engineer · Full-Stack Developer · AI/ML Enthusiast

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PROFESSIONAL SUMMARY

Associate Software Engineer at Fentons IT with 1+ year of industry experience building scalable full-stack applications for enterprise clients. Final-year BSc (Hons) Information Technology student at SLIIT, specializing in modern web technologies including, Next.js, React 18, Node.js, Express.js, and MySQL. Proven track record delivering production-grade systems with CI/CD pipelines, automated testing, and AWS deployments. Complementary expertise in AI/ML (XGBoost, PyTorch, OpenCV) and a natural team leader - Scout instructor, national jamboree representative, and competitive swimmer.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), Java, Python, PHP, SQL, Kotlin, HTML5/CSS3, C++, R

Frontend: Next.js 15, React 18, Redux, Tailwind CSS, Responsive Design, Cross-browser Compatibility

Backend: Node.js, Express.js, Spring Boot, REST APIs, Microservices, Firebase, FastAPI

Databases: MySQL, MongoDB, PostgreSQL, NoSQL, Database Design & Normalization

Cloud & DevOps: AWS (EC2, S3), Docker, GitHub Actions (CI/CD), Git, Gitflow, Linux

Testing: Jest, React Testing Library, Postman, API Testing, Unit Testing, Integration Testing

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, COLMAP, Hugging Face, NLP

Practices: Agile/Scrum, SDLC, OOP, UML, Code Review, Data Structures & Algorithms, Figma

WORK EXPERIENCE

Associate Software Engineer · Fentons IT

2024 – Present

- Architected and delivered full-stack features for an enterprise Construction Project Management (CPM) platform serving 5+ enterprise clients, using Next.js 15, React 18, and Express.js within a 6-person Agile team.
- Engineered RESTful APIs integrated with MySQL, implementing JWT-based authentication, role-based access control, and secure session management - reducing unauthorized access incidents to zero post-launch.
- Built multi-level reporting dashboards and financial monitoring modules that consolidated data from 4+ procurement workflows, cutting manual reporting time by approximately 35%.
- Established GitHub Actions CI/CD pipelines for automated testing and deployment to AWS EC2, reducing deployment time from manual 2-hour process to under 15 minutes.
- Participated in weekly code reviews and sprint retrospectives, contributing to a 20% improvement in sprint velocity over 3 months through improved task estimation practices.
- Wrote unit and integration tests using Jest and Postman to maintain >80% code coverage across API endpoints, ensuring reliability for production enterprise deployments.

EDUCATION

BSc (Hons) Information Technology - Specialization in Information Technology · SLIIT

2022 – Present

Sri Lanka Institute of Information Technology · Final Year · Expected Graduation: 2026

GCE Advanced Level - Physical Science · Royal College Colombo-07

2008 – 2021

Results: 1C & 2S

PROJECTS

CPM System - Enterprise Construction Project Management Platform *Next.js 15 · React 18 · Express.js · MySQL · AWS · Docker*

- Designed and built an enterprise-grade web platform managing complex construction projects for 5+ client organizations, featuring role-based access control, real-time collaboration, and multi-level reporting.
- Delivered financial monitoring, procurement tracking, and compliance dashboards that reduced manual data entry by ~40% for project managers across client teams.
- Containerized the application with Docker and deployed to AWS EC2 with automated CI/CD via GitHub Actions, achieving 99.5% uptime in production.

Smart Construction Intelligence System *FastAPI · Python · XGBoost · Flutter · Scikit-learn · REST APIs*

- Led a team of 4 to build an AI-powered microservices platform that predicts construction project costs, resource timelines, and material quality - reducing estimation effort by 60% compared to manual processes.
- Architected 4 specialized backend microservices with REST API communication, integrating XGBoost regression models that achieved 89% prediction accuracy on the construction cost dataset.
- Wrote integration tests with Postman and automated test suites validating all 4 services, ensuring data integrity across the prediction pipeline.

Sustainable Community Management System *Node.js · React.js · Express.js · MongoDB · Firebase · Redux*

- Developed a full-stack community platform supporting 50+ active users with real-time event scheduling, resource management, and interactive dashboards - reducing event coordination time by 30%.
- Implemented Redux state management and Tailwind CSS for a fully responsive, cross-device UI with cross-browser compatibility across Chrome, Firefox, and Safari.
- Deployed with Firebase real-time database integration, achieving sub-200ms query response times for dashboard data loads.

Tinkerbelle Vehicle Rental System *PHP · JavaScript · MySQL · HTML5/CSS3*

- Built a complete rental platform with user authentication, concurrent booking management, and vehicle availability checks - supporting 20+ simultaneous users in testing.
- Designed and normalized a MySQL schema with 8 tables, enforcing referential integrity and optimizing queries to reduce average page load time by 25%.

Hotel Management System (In Progress) *MERN Stack · ML · Docker*

- Building a MERN-stack hotel management platform with ML-driven dynamic pricing that adjusts room rates based on occupancy and seasonality patterns, targeting 15% revenue optimization.

CERTIFICATIONS & LEARNING

- Meta Front-End Developer Professional Certificate - Coursera (In Progress)
- AWS Cloud Practitioner Essentials - AWS Training (Completed Self-Study)
- JavaScript Algorithms & Data Structures - freeCodeCamp

LEADERSHIP & EXTRACURRICULAR

Scout Leader / Assistant Instructor · 42nd Colombo Scout Group, Royal College

2017 – Present

- Mentored 30+ young scouts as Assistant Instructor for the Gold Troop, designing training programmes that improved patrol performance scores by 40% over two seasons.
- Represented the troop at 6 national and international events including the 9th National Jamboree (2016) and Colombo Centenary International Scout Jamboree (2014), coordinating logistics for groups of 15–25 participants.
- Led Hawks Patrol to championship victory at the 27th Lt. Col. M.K.J. Cantlay Challenge Shield (2016), competing against 20+ patrols from Colombo district.

Competitive Swimming · Royal College Inter House Aquatic Meet

- 1st Place - U/17 50m Backstroke (2017)
- 1st Place - U/17 50m Breaststroke (2017)

REFERENCES

Dr. Samantha Rajapaksha

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